

Business Calculus (MATH 2350)

Summer 2018

Monday, 06/25/2018

Quiz 8: Section 13.1, 13.2

Name (Print): _____

Score: _____ / 100

Section 13.1

1. 10 points Find $\int \frac{2}{x} dx$

$$\begin{aligned}\int \frac{2}{x} dx &= 2 \int \frac{1}{x} dx \\ &= 2\ln(|x|) + c\end{aligned}$$

2. 10 points Find $\int 4t^3 + 3t + 3 dt$

$$\begin{aligned}\int 4t^3 + 3t + 3 dt &= 4 \int t^3 dt + 3 \int t dt + \int 3 dt \\ &= 4\left(\frac{1}{4}t^4\right) + 3\left(\frac{1}{2}t^2\right) + 3t + c \\ &= t^4 + \frac{3}{2}t^2 + 3t + c\end{aligned}$$

3. 10 points If the marginal profit of producing x surfboards is given by $P'(x) = 40 - x$, and $P(10) = 20$ where $P(x)$ is the profit in dollars, find $P(x)$.

First, we find $\int P'(x)dx$:

$$\int 40 - x \, dx = 40x - \frac{1}{2}x^2 + c.$$

Then, we solve for c :

$$\begin{aligned} P(10) &= 20 \\ 40(10) - \frac{1}{2}(10)^2 + c &= 20 \\ 400 - 50 + c &= 20 \\ c &= 20 - 400 + 50 \\ c &= -330. \end{aligned}$$

So, $P(x) = 40x - \frac{1}{2}x^2 - 330$.

4. 10 points Find $\int e^x + 2x^4 + \frac{2}{x} dx$.

$$\begin{aligned} \int e^x + 2x^4 + \frac{2}{x} dx &= \int e^x dx + 2 \int x^4 dx + 2 \int \frac{1}{x} dx \\ &= e^x + 2\left(\frac{1}{5}x^5\right) + 2\ln(|x|) + c \\ &= e^x + \frac{2}{5}x^5 + 2\ln(|x|) + c \end{aligned}$$

5. 15 points (bonus) If the marginal profit of producing x items is given by $P'(x) = 40 - x$, and $P(x_0) = n$ (where $P(x)$ is the profit in dollars of producing x units, x_0 is some fixed number of units, and n is some fixed number of dollars of profit), find a general formula for $P(x)$.

(Hint: Look at problem 3 and change your values accordingly)

First, we find $\int P'(x)dx$:

$$\int 40 - x \, dx = 40x - \frac{1}{2}x^2 + c.$$

Then, we solve for c :

$$\begin{aligned} P(x_0) &= n \\ 40(x_0) - \frac{1}{2}(x_0)^2 + c &= n \\ c &= n - 40x_0 + \frac{x_0^2}{2} \end{aligned}$$

So, $P(x) = 40x - \frac{1}{2}x^2 + (n - 40x_0 + \frac{x_0^2}{2})$.

Section 13.2

1. In each problem, find $\int f(x) dx$.

(a) 10 points $f(x) = 24x(3x^2 + 2)^3$

(Hint: If du is too large or small, consider scaling du)

Let $u = 3x^2 + 2$. Then, $du = 6x dx$ and $4du = 24x dx$. We now substitute and solve:

$$\begin{aligned}\int 24x(3x^2 + 2)^3 dx &= \int 4u^3 du \\ &= 4\left(\frac{1}{4}u^4\right) + c \\ &= u^4 + c \\ &= (3x^2 + 2)^4 + c.\end{aligned}$$

(b) 10 points $f(x) = \frac{x^2}{x^3 + 1}$

Let $u = x^3 + 1$. Then, $du = 3x^2 dx$ and $\frac{1}{3}du = x^2 dx$. We now substitute and solve:

$$\begin{aligned}\int \frac{x^2}{x^3 + 1} dx &= \int \frac{1}{3} \left(\frac{1}{u}\right) du \\ &= \frac{1}{3} \ln(|u|) + c \\ &= \frac{1}{3} \ln(|x^3 + 1|) + c.\end{aligned}$$

- (c) 10 points $f(x) = (x - 1)(x + 1)^5$
(Hint: If $u = f(x)$, then $f(x) - 2 = u - 2$)

Let $u = x + 1$. Then, $du = dx$ and $u - 2 = x - 1$. We now substitute and solve:

$$\begin{aligned}\int (x - 1)(x + 1)^5 dx &= \int (u - 2)(u)^5 du \\ &= \int u^6 - 2u^5 du \\ &= \frac{1}{7}u^7 - \frac{2}{6}u^6 + c \\ &= \frac{1}{7}(x + 1)^7 - \frac{2}{6}(x + 1)^6 + c \\ &= (x + 1)^6 \left(\frac{1}{7}(x + 1) - \frac{2}{6} \right) + c \\ &= (x + 1)^6 \left(\frac{1}{7}x - \frac{4}{21} \right) + c.\end{aligned}$$

- (d) 10 points $f(x) = e^{2x} + 2x + 2$

Let $u = 2x$. Then, $du = 2dx$ and $\frac{1}{2}du = dx$. We now substitute and solve:

$$\begin{aligned}\int e^{2x} + 2x + 2 dx &= \int \frac{1}{2} (e^u + u + 2) du \\ &= \frac{1}{2} \left(\int e^u du + \int u du + \int 2 du \right) \\ &= \frac{1}{2} \left(e^u + \frac{1}{2}u^2 + 2u \right) + c \\ &= \frac{1}{2}e^u + \frac{1}{4}u^2 + u + c \\ &= \frac{1}{2}e^{2x} + \frac{1}{4}(2x)^2 + 2x + c \\ &= \frac{1}{2}e^{2x} + x^2 + 2x + c.\end{aligned}$$

2. 10 points Find $\int 2xe^{x^2} dx$.

Let $u = x^2$. Then, $du = 2x dx$. We now substitute and solve:

$$\begin{aligned}\int 2xe^{x^2} dx &= \int e^u du \\ &= e^u + c \\ &= e^{x^2} + c.\end{aligned}$$

3. 10 points If the marginal cost for producing x of a certain item is given by $C'(x) = 4(x+10)^3$, and the fixed cost is $C(0) = 2000$ (in dollars), find the cost function $C(x)$.

First, we solve $\int C'(x) dx$. Let $u = x + 10$. Then $du = dx$, and we may proceed to substitute and solve:

$$\int 4(x+10)^3 dx = \int 4(u)^3 du = u^4 + c = (x+10)^4 + c.$$

Now, solve for c :

$$\begin{aligned}C(0) &= 2000 \\ ((0) + 10)^4 + c &= 2000 \\ 10^4 + c &= 2000 \\ c &= 2000 - 10^4.\end{aligned}$$

So, $C(x) = (x+10)^4 + (2000 - 10^4)$.

4. 20 points (bonus) Find $\int \frac{20x^3 + 2x + 1}{5x^4 + x^2 + x} \ln(10x^4 + 2x^2 + 2x) dx$.

Let $u = \ln(10x^4 + 2x^2 + 2x)$. Then,

$$\begin{aligned} du &= \frac{40x^3 + 4x + 2}{10x^4 + 2x^2 + 2x} dx \\ &= \frac{2(20x^3 + 2x + 1)}{2(5x^4 + x^2 + x)} dx \\ &= \frac{20x^3 + 2x + 1}{5x^4 + x^2 + x} dx. \end{aligned}$$

We now substitute and solve:

$$\begin{aligned} \int \frac{20x^3 + 2x + 1}{5x^4 + x^2 + x} \ln(10x^4 + 2x^2 + 2x) dx &= \int u du \\ &= \frac{1}{2} u^2 + c \\ &= \frac{1}{2} (\ln(10x^4 + 2x^2 + 2x))^2 + c. \end{aligned}$$